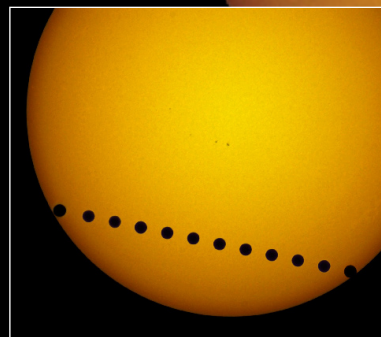


■ ALIGNMENTS

The planets and the Moon are always seen within the same band of sky. So it is not surprising that two or more are often seen close together at the same time. Such an alignment is called a conjunction. Should one object obstruct the view of another, more distant object, it is known as an occultation. The Moon regularly occults background stars. When the Moon or a planet passes in front of a star, it is temporarily lost from view. In the case of the Moon, which has no atmosphere, the disappearance and later reappearance of the star happens instantly. A grazing occultation is when the star skims the upper or lower limb of the Moon. In this case, the occulted body is lost from view only as it disappears behind lunar mountain peaks along the limb. An occultation of a bright planet by the Moon occurs 10 or 11 times a year. Occultations of one planet by another—for example, an occultation of Jupiter by Venus—occur only a few times a century.

OTHER CELESTIAL PHENOMENA

When the Moon obstructs our view of the Sun it is also, technically, an occultation, but this phenomenon is better known as an eclipse. During a solar eclipse, the Moon stops the Sun's light from reaching Earth. When the Moon moves into Earth's shadow, we witness a lunar eclipse.



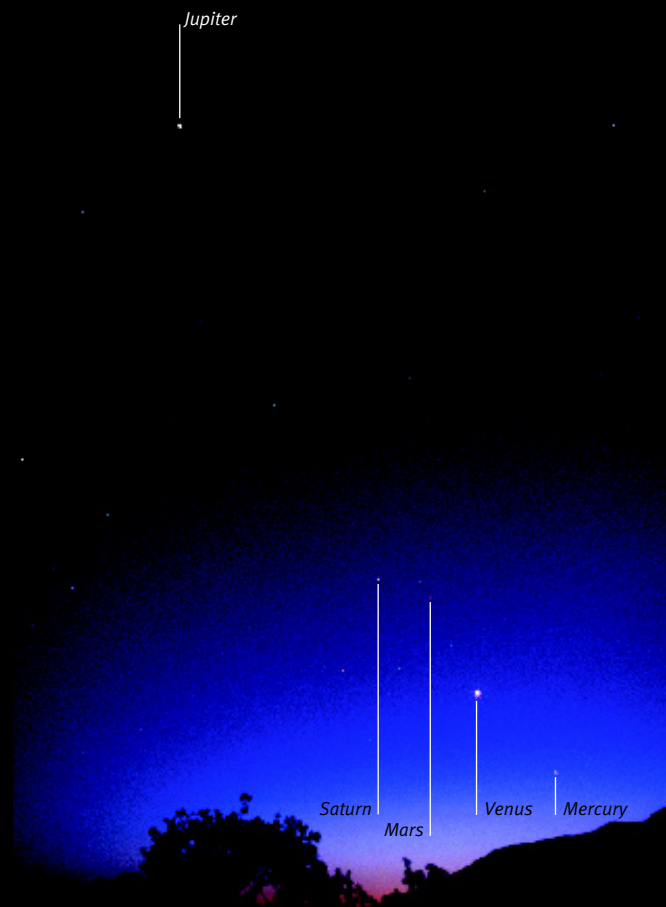
Path across the Sun

This composite image of the 2004 transit of Venus charts the progress of the planet across the Sun's disk. It took more than six hours for the planet to cross the Sun. Transits of Venus occur in pairs, and the partner to this one is due in 2012. The next pair is due in 2117 and 2125.



Occultation of Jupiter

This picture shows the January 2002 occultation of Jupiter, which is about to disappear from view behind the Moon. The two objects are along the same line of sight from Earth. Such an alignment often lasts long enough for an occultation to occur each lunar month until the two move out of alignment.



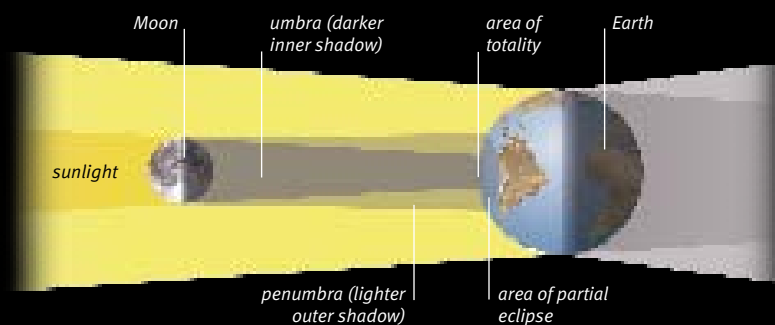
Planetary alignment

In April 2002, soon after the Sun had set, a line of five planets could be seen with the naked eye. Nearest to the horizon was Mercury (bottom right), above and to its left was brighter Venus, and up and to the left again were Mars and Saturn. Bright Jupiter (top left) completed the view.

MECHANICS OF ECLIPSES

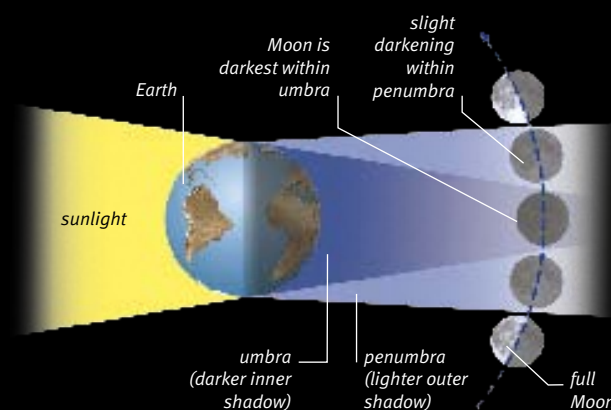
The Moon is 400 times smaller than the Sun and 400 times closer to Earth, so when the three bodies are directly aligned, the Moon's disk blocks the Sun from view. This is a solar eclipse. The shadow cast by the Moon falls on Earth, and within the shadow,

day turns to night as darkness falls. A total solar eclipse typically lasts for 3–4 minutes. In a lunar eclipse, when the Moon is in Earth's shadow, totality can last up to 1 hour 47 min. There are on average four eclipses each year—two solar and two lunar.



Eclipse of the Sun

When the Moon is directly between the Sun and Earth, it blots out the Sun and casts a shadow on Earth. Anyone within the umbra, the darker inner shadow, sees a total eclipse. Outside of this is a wider area of lighter shadow, the penumbra, from within which a partial eclipse is seen. An annular eclipse occurs when the Moon is farther from Earth than average, and its disk is too small to cover the Sun's disk totally.



Eclipse of the Moon

The Sun, Earth, and Moon are aligned, and Earth stops sunlight from reaching the Moon. When the whole Moon is in the umbra, it receives no light and is totally eclipsed. When only a portion of the Moon passes through the umbra, a partial eclipse can be seen.